

CO2 - AT(2)

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:25

Duration : 1 Hour

Design Task

Code of Conduct:

I Om Sai Reddy T 192372252 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

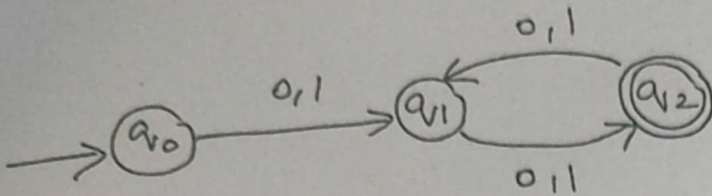
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Om Sai Reddy T

SIMATS ENGINEERING ASSESSMENT TOOL

1. Design the regular expression for the language represented by the given automata



2. Consider the following grammar

$S \rightarrow aSc|T|U|e$

$T \rightarrow bTc|e$

$U \rightarrow aUb|e$

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbc

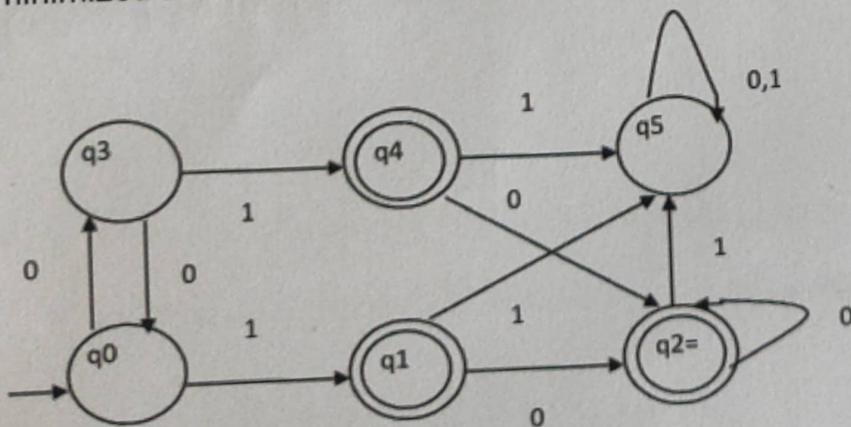
C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

3. Design a non deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression 00^*1+11^*0

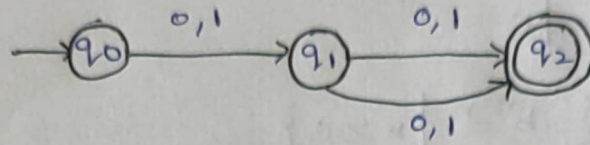
4. i) Show that if L_1 and L_2 are regular then (L_1+L_2) also regular

ii) Is the language $\{a^p | p \text{ is prime}\}$ regular? Justify

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



Design the regular Expression for the language represented by the given Automata.



A) Step ①:-

q_0 is the initial state.

q_2 is the final state.

Observation:-

To reach the final state q_2 :-

① 1st Symbol :- $q_0 \rightarrow q_1$

② 2nd Symbol :- $q_1 \rightarrow q_2$

③ Every additional accepts all strings of even length over $\{0,1\}$

Regular Expression:-

$$(0+1)(0+1)^*$$

Accepted string:-

00
01
10
11

② Consider the following Grammar:-

$$S \rightarrow asc|T|U|e$$

$$T \rightarrow bTc|e$$

$$U \rightarrow aub|e$$

what is the language defined by the Grammar

Grammar

A)

For T
 $\Rightarrow * =$

For S
 $\Rightarrow * =$

$$T \rightarrow bTc \mid \epsilon$$

$$\therefore L(T) = \{ b^n c^n \mid n \geq 0 \}$$

ex:- $\epsilon, bc, bbcc, bbbccc, \dots$

For U

$$U \rightarrow aUb \mid \epsilon$$

$$\therefore L(U) = \{ a^n b^n \mid n \geq 0 \}$$

ex:- $\epsilon, ab, aabb, aaabbbb, \dots$

From ①:- Using T

$$S \Rightarrow a^k T c^k$$

Since T provides $b^n c^n$

$$a^k b^n c^{n+k}$$

Thus:-
$$L_1 = \{ a^k b^n c^{n+k} \mid k, n \geq 0 \}$$

Form ②:- Using 'U'

$$S \Rightarrow a^k U c^k$$

$$a^{k+n} b^n c^k$$

$$L_2 \rightarrow \{ a^{k+n} b^n c^k \mid k, n \geq 0 \}$$

$$\therefore L(C-1) = \{ a^k b^n c^{n+k} \mid k, n \geq 0 \} \cup \{ a^{k+n} b^n c^k \mid k, n \geq 0 \}$$

Grammar "aaabbc"

$$S \rightarrow asc$$

$$S \rightarrow auc \quad S \rightarrow u$$

$$S \rightarrow aaabbc$$

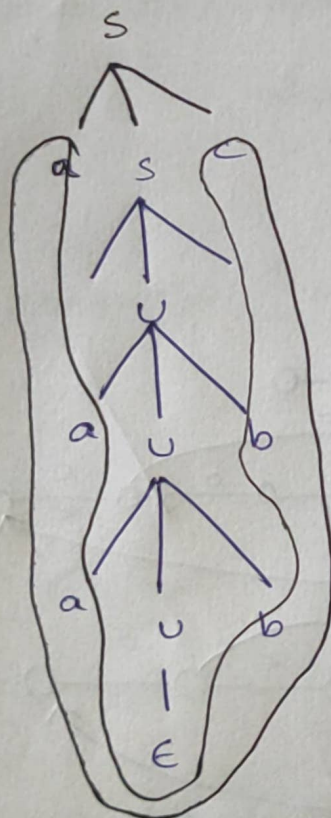
$$U \rightarrow \epsilon$$

$$S \rightarrow aaUbc \quad U \rightarrow aub$$

$$S \rightarrow aaqUbbc \quad U \rightarrow aub$$

$\therefore$ string is accepted

Parse Tree :-
==* ==* ==



$\therefore$ string "aaabbc"

(c) Is the Grammar Ambiguous?

Derivation (1)

$$S \rightarrow asc$$

$$S \rightarrow auc \quad S \rightarrow u$$

$$S \rightarrow aaUbc \quad U \rightarrow aub$$

$$S \rightarrow aaqubbc \quad U \rightarrow aub$$

Derivation (2) :-

$$S \rightarrow asc$$

$$S \rightarrow auc \quad S \rightarrow u$$

$$S \rightarrow aaUbc \quad U \rightarrow aub$$

$$S \rightarrow aaTc \quad U \rightarrow T$$

$$S \rightarrow aaabbc \quad U \rightarrow \epsilon$$

$$S \rightarrow aaabTC \quad T \rightarrow \epsilon$$

$$S \rightarrow aaabbc \quad T \rightarrow \epsilon$$

$\therefore$ The Grammar is Ambiguous, because it having two derivation in LMD.

(3)

Design a non-deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression.

$$00^*1 + 11^*0?$$

$$00^*1 + 11^*0$$

$$0 \rightarrow 0 \xrightarrow{0} 0$$

$$00^* \rightarrow 0 \xrightarrow{0} 0 \xrightarrow{\epsilon} 0 \xrightarrow{0} 0$$

$$00^* \rightarrow 0 \xrightarrow{\epsilon} 0 \xrightarrow{0} 0 \xrightarrow{\epsilon} 0 \xrightarrow{0} 0 \xrightarrow{\epsilon} 0$$

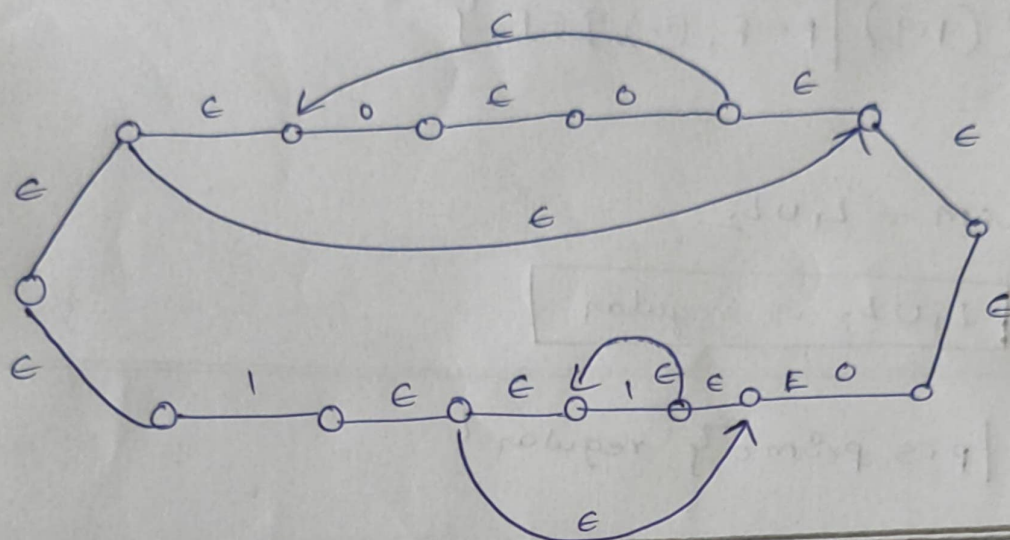
$$00^*1 \rightarrow 0 \xrightarrow{\epsilon} 0 \xrightarrow{0} 0 \xrightarrow{\epsilon} 0 \xrightarrow{0} 0 \xrightarrow{\epsilon} 0 \xrightarrow{1} 0$$

$$11^* \rightarrow 1 \xrightarrow{\epsilon} 1 \xrightarrow{1} 1 \xrightarrow{\epsilon} 1 \xrightarrow{1} 1 \xrightarrow{\epsilon} 1$$

$$11^*0 \rightarrow 1 \xrightarrow{\epsilon} 1 \xrightarrow{1} 1 \xrightarrow{\epsilon} 1 \xrightarrow{1} 1 \xrightarrow{\epsilon} 1 \xrightarrow{0} 0$$

(3)

$T \rightarrow 0^*1 + 11^*0 \rightarrow$
 $T \rightarrow \epsilon$



(4) I) Show that if L_1 and L_2 are regular then $(L_1 + L_2)$ also regular.

(i) is the language $\{a^p \mid p \text{ is prime}\}$ regular?

A) Given :-

Suppose :-

$L_1 \& L_2$

$M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ and $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$

$L(M_1) = L_1 \quad \& \quad L(M_2) = L_2$

Now DFA :-

$M = (Q_1 \& Q_2, \Sigma, \delta, (q_1, q_2), F)$

Where :-

$\delta((p, q), a) = (\delta_1(p, a), \delta_2(q, a))$

final states :-
 $= * = * = * =$

$$F = \{ (p, q) \mid p \in F_1, (or) q \in F_2 \}$$

Hence :-

$$LCM = L_1 \cup L_2$$

$L_1 \cup L_2$ is regular

(ii) $\{a^p \mid p \text{ is prime}\}$ regular?

$$L = \{a^p \mid p \text{ is prime}\}$$

Step ① :-

Assume L is regular

$$L = \{a^p \mid p \text{ is prime}\}$$

$w = a^p$ belongs to L .

$$w = xyz$$

Where :-

$$|xy| \leq m$$

$$|y| > 0$$

$$|y| = k$$

$$1 \leq k \leq m$$

Step ② :-

$$s = p+1$$

$$xy^{p+1}z$$

$$p+p^k$$

Factorizing
 $= * = *$

$$p + pk = p(1+k)$$

$$p > 1$$

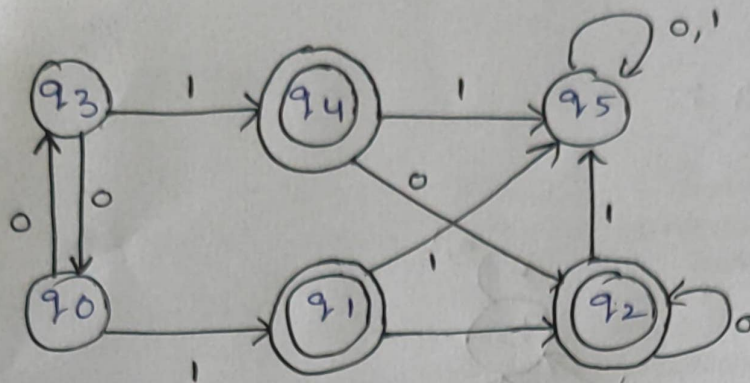
$$1+k > 1$$

$$p(1+k)$$

$$xy^{p+1}z \in L$$

$\therefore L = \{a^p \mid p \text{ is prime}\}$ is not regular

Minimize the given DFA ?



Step ① :-

Transition table :-

State / I/P	0	1
q ₀	q ₃	q ₁
q ₁	q ₀	q ₄
q ₂	q ₂	q ₅
q ₃	q ₂	q ₅
q ₄	q ₂	q ₅
q ₅	q ₅	q ₅

Iteration 0 :-

Initial = { q₀, q₃, q₂ }

Final = { q₁, q₂, q₄ }

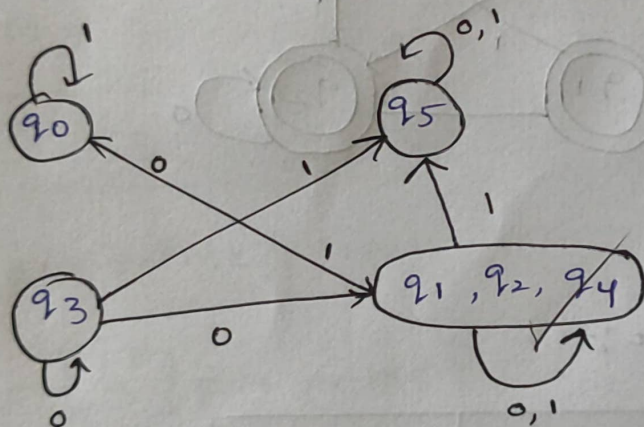
I₁ :- I/P :- '0'

{ q₅ } { q₀ } { q₃ } { q₁, q₂, q₄ }

I

I_2 :- $\{q_5\}$ $\{q_0\}$ $\{q_3\}$ $\{q_1, q_2, q_4\}$

Minimized DFA :-
= * = * = * =



SIMATS ENGINEERING ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

91/100